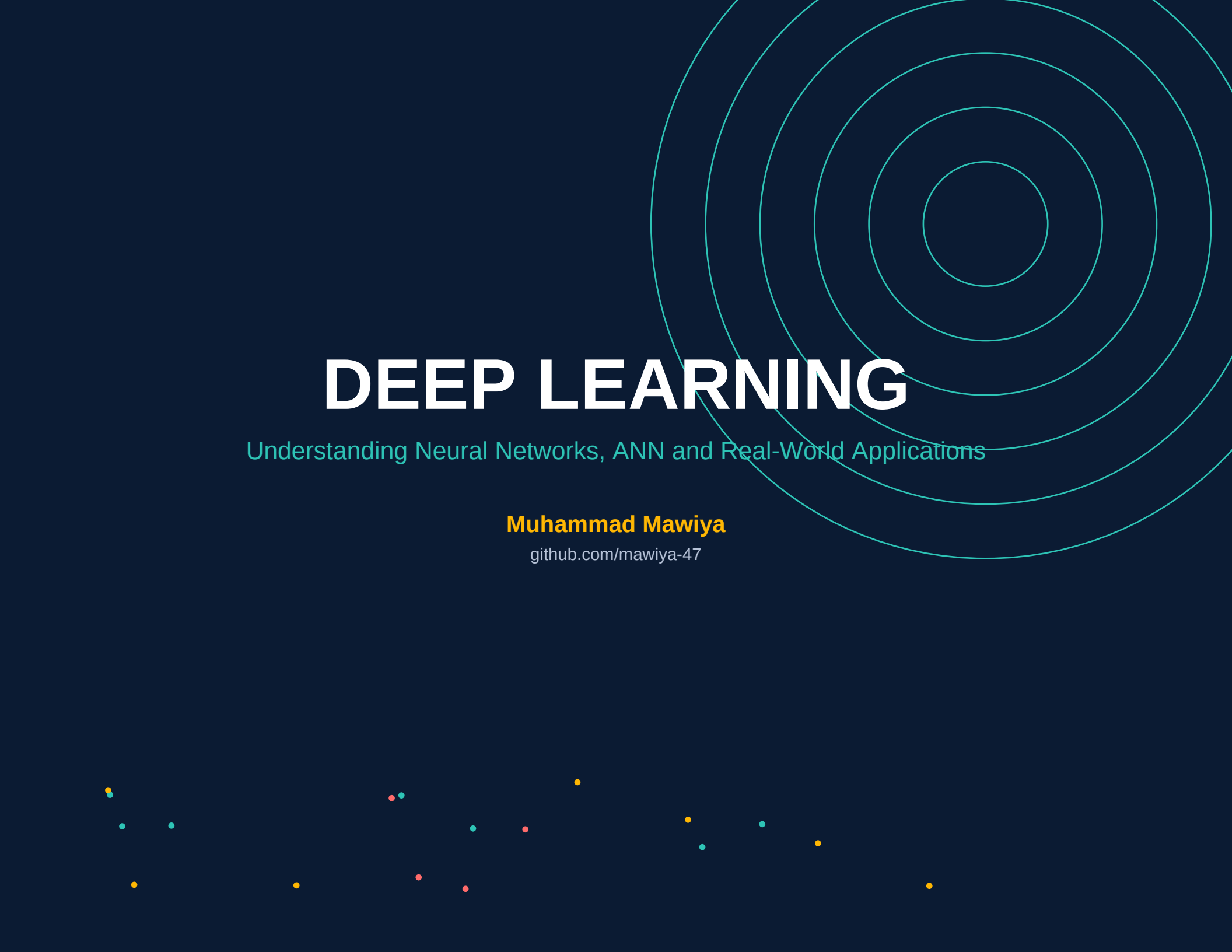


The background features a dark blue gradient. In the top right corner, there are several concentric circles in a light teal color. In the bottom left corner, there are several small dots in yellow, teal, and red.

DEEP LEARNING

Understanding Neural Networks, ANN and Real-World Applications

Muhammad Mawiya

github.com/mawiya-47

Machine Learning vs Deep Learning

Machine Learning

A subset of AI that uses statistical methods, enabling machines to learn patterns and improve with experience.

Deep Learning

A subset of Machine Learning that uses multi-layer neural networks to automatically learn representations directly from raw data.

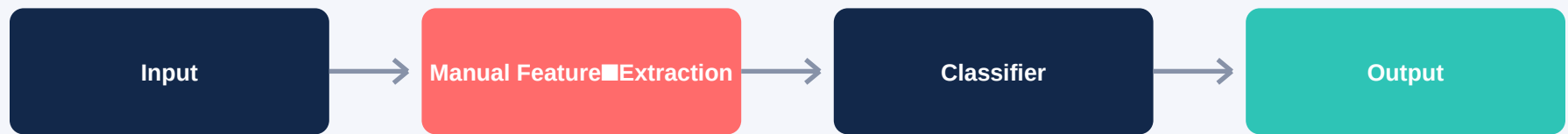
In short: ML needs human-crafted features. DL builds its own features from data.

ML vs DL — Side by Side

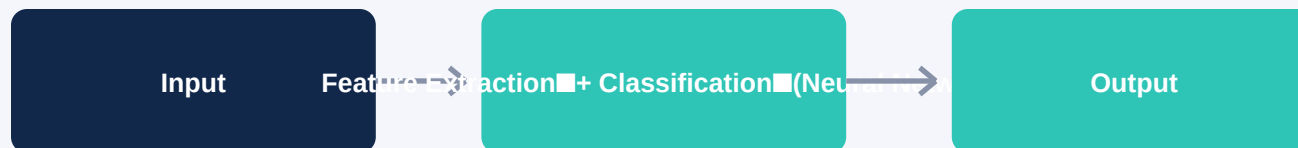
Aspect	Machine Learning	Deep Learning
Approach	Finds patterns using a variety of algorithms	Learns representations automatically from raw data
Feature Engineering	Manual extraction & selection	Automatic — no manual feature design
Versatility	Regression, classification, clustering, anomaly detection	Excels at vision, speech, and NLP tasks
Data Requirement	Works fine with smaller datasets	Needs large datasets to perform well
Compute Intensity	Can run on standard machines	Usually needs GPUs / high-end hardware

The Core Difference: Feature Engineering

Machine Learning

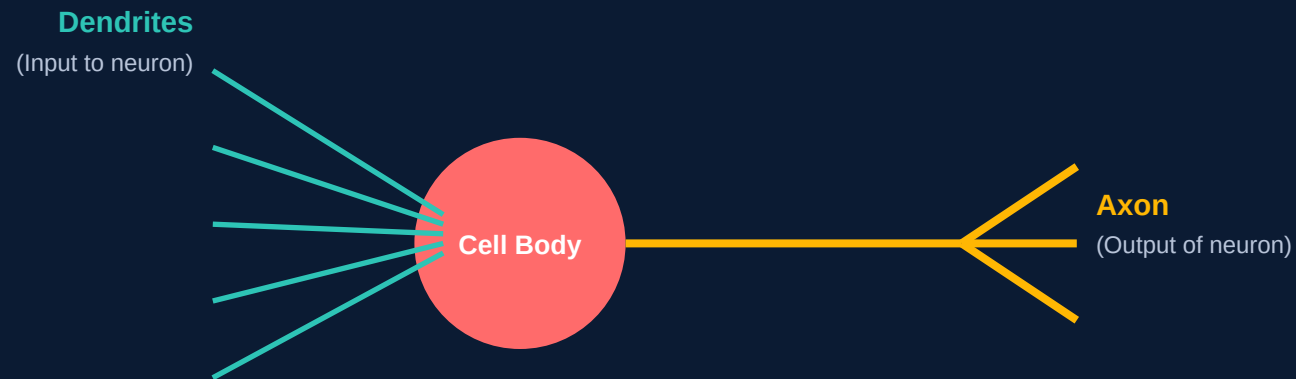


Deep Learning



Deep Learning collapses feature extraction and classification into a single learned pipeline.

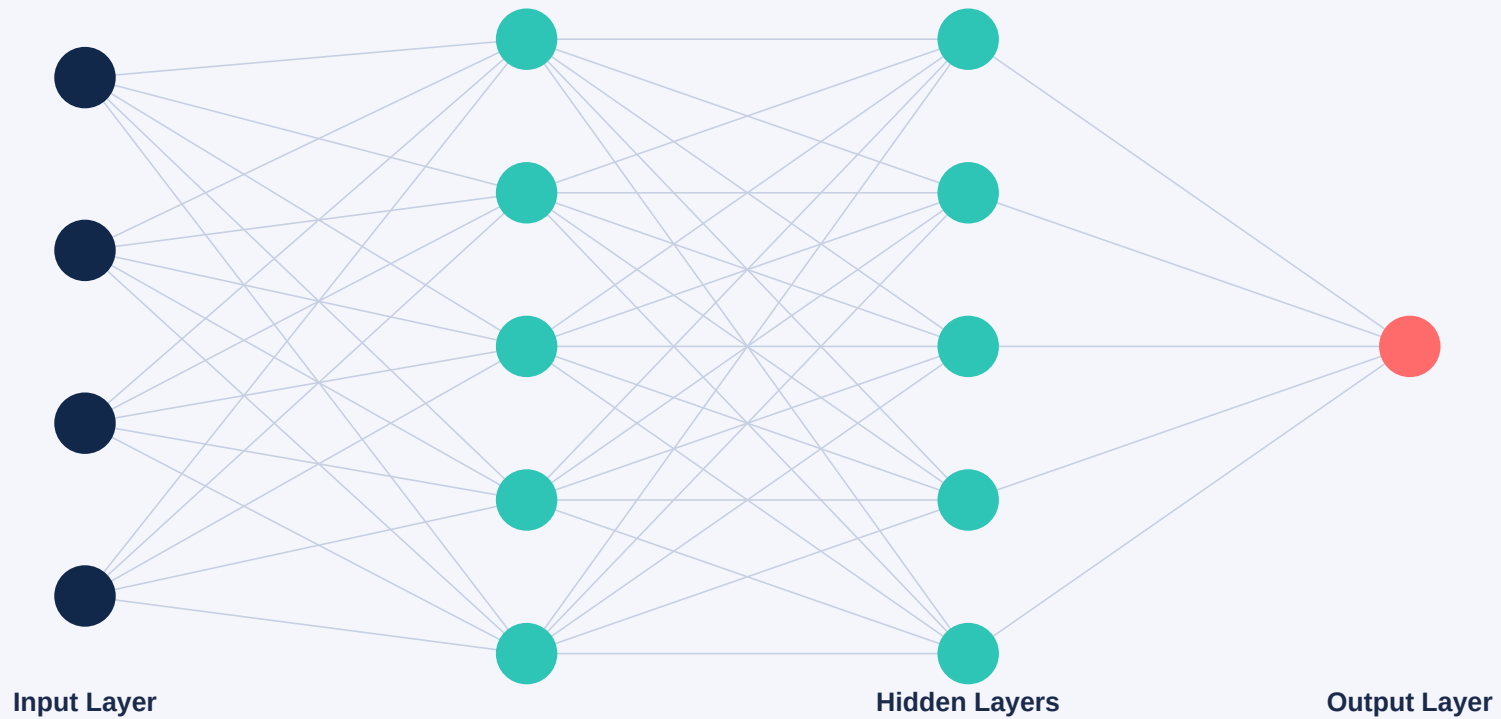
Where It All Started: The Biological Neuron



Since the 1950s, scientists have tried to mimic how a biological neuron processes information — this idea eventually gave rise to Artificial Neural Networks.

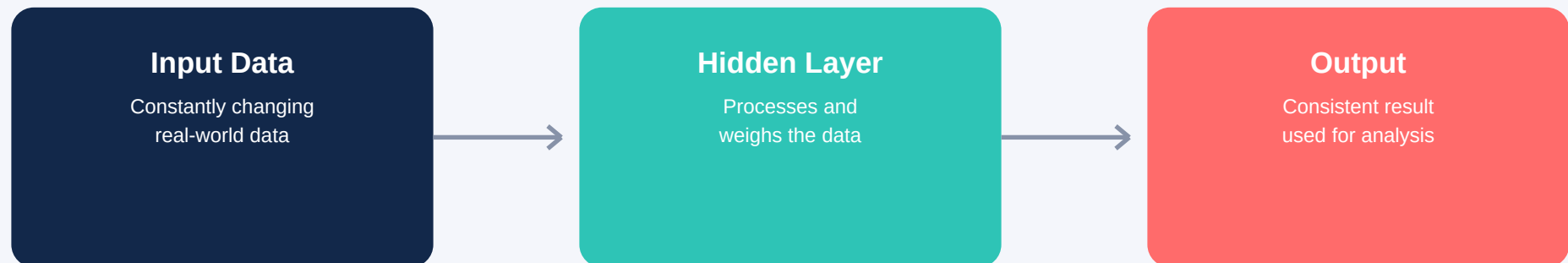
Artificial Neural Network (ANN)

A system of hardware/software patterned after neurons in the human brain.



How a Neural Network Learns

Like the human brain, a neural network constantly adjusts itself based on new input.



Use Cases for Neural Networks

Pattern Recognition

Identifies patterns in sales/customer data to support informed business decisions.

Forecasting

Brings an added layer of sophistication to business and financial forecasting.

Credit Worthiness

Predicts bankruptcy risk, helping banks assess loan applications.

Valuation

Improves the accuracy of valuation models across industries.

Applications of ANN in the Real World

● Handwriting Recognition

Converts handwritten characters into digital text the system can recognize.

● Stock Exchange Prediction

Examines many market factors to help predict daily prices for brokers.

● Travelling Salesman Problem

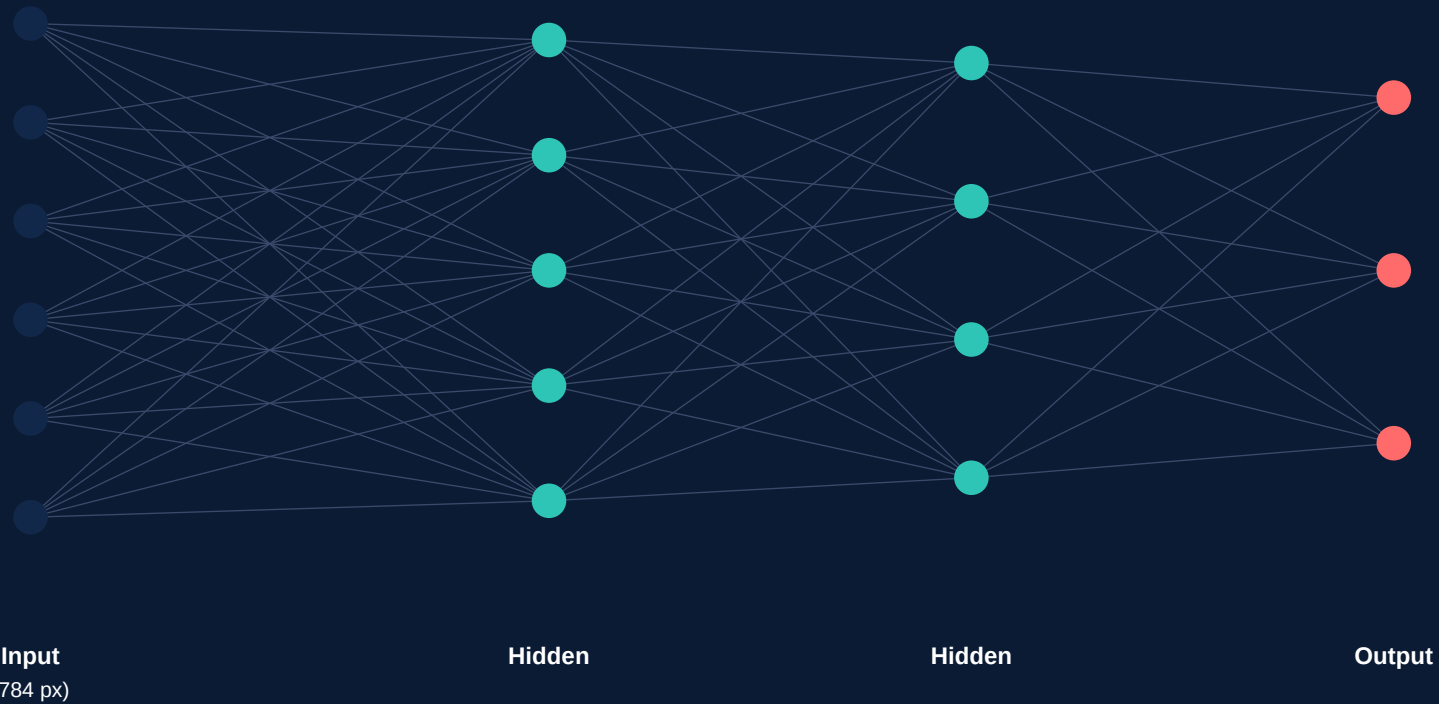
Finds the optimal route between cities for maximum revenue, minimum cost.

● Image Compression

Stores, encrypts and recreates images efficiently using learned representations.

Forward Propagation — A Quick Look

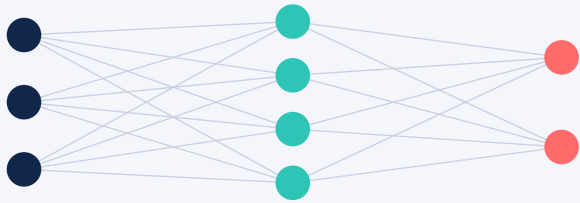
A 28x28 pixel image (784 inputs) is fed through weighted connections, processed layer by layer, and mapped to an output class.



Single vs Multi Hidden-Layer Networks

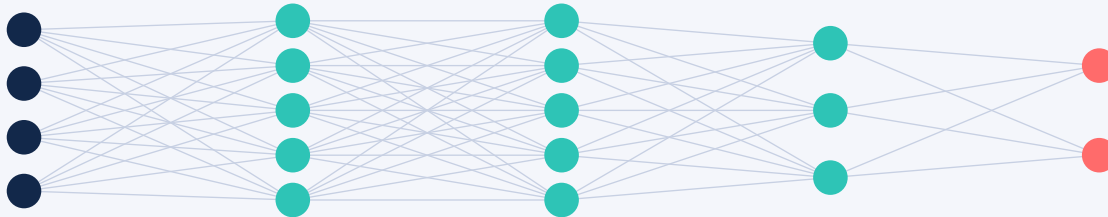
Single Hidden-Layer Network

One input layer, one hidden layer, one output layer.



Multi Hidden-Layer Network (Deep Network)

One input layer, several hidden layers, one output layer — this depth is what makes it 'deep' learning.



Thank You

Deep Learning — Concepts, Neural Networks & Applications

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